

CORNELIUS MULYOKELA

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EDUCATION

Stetson University

B.S. Computer Science and Applied Mathematics (Data Science Concentration)
Stetson Merit Scholarship Recipient | **3.81/4.0** GPA

DeLand, FL
Expected May 2027

SKILLS

Languages: Python, Java, C#, C++, JavaScript, HTML, CSS, SQL, MATLAB, R

Technologies: Kafka, Spark, Airflow, PostgreSQL, ETL & data pipelines, AWS (S3, EC2), Flask, Tableau, MySQL, NumPy, Docker, Linux, Bootstrap, React, Node.js, SQLite

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Software Development, Database Systems, Distributed Systems, Data Mining, Machine Learning, Linear Algebra, Probability and Statistics, Statistical Modeling, Optimization

PROFESSIONAL EXPERIENCE

University of Pennsylvania Perelman School of Medicine

Software Engineering & Medical Research Intern

Philadelphia, PA
May 2025 – August 2025

- Developed a **scalable Python-based MRI Spectroscopy analysis tool** replacing legacy **MATLAB-based** workflows, improving research efficiency by **40%**
- Designed **imaging pipelines** that processed **large-scale unstructured medical data**, reducing analysis time from hours to minutes
- Applied **statistical modeling** and **algorithms** to identify patterns in brain imaging data, generating actionable insights for clinical researchers

ProWeb Ltd

Software Engineering Intern

Remote
May 2024 – August 2024

- Provided hands-on technical support to over **200 users**, diagnosing and resolving system-level issues across cloud and local environments, reducing downtime by **65%**
- Implemented **SQL-based** reporting to track **KPIs**, enabling **data-driven decision-making** for business operations
- Created internal documentation and **toolkits** to improve support handoffs and knowledge retention across the engineering team

Stetson University Center for Public Opinion Research

Data Analyst

DeLand, FL
Aug 2023 – Dec 2023

- Designed and executed statistical analyses on survey data from **43,000+ participants** using **Qualtrics, R, and Excel**
- Built **regression** models and **performed hypothesis** testing to inform **public funding decisions**
- Translated **complex datasets** into insights using **Excel and R**, influencing strategic municipal investment decisions

Math & Computer Science Department - Stetson University

Research Assistant

DeLand, FL
September 2025 – Present

- Collaborated with faculty on computational research projects involving **algorithm development, data preprocessing, and experimental analysis using Python and R**
- Developed Python scripts to automate data preprocessing and analysis workflows, reducing manual processing time by 40%.
- Assisted **50+ students** in **mathematics and programming** coursework, increasing course comprehension by **25%**

PERSONAL PROJECTS

MRI Frequency Shift Tool (UPenn-FIRE) | Python, FFT, Docker, Siemens FIRE

- Developed a **Python-based** spectral preprocessing tool using **FFT** to analyze MRI spectroscopy data
- Integrated the tool with **Siemens FIRE platform workflows**, improving MRI signal preprocessing **efficiency by 80%**
- Built **Docker** containers to package the pipeline and ensure reproducible execution across **computing environments**

Real-Time Environmental Streaming Pipeline | Python, Kafka, Spark, Airflow, PostgreSQL

- Built a **distributed streaming pipeline** that simulates environmental sensor data and processes it with **Kafka and Spark** Structured Streaming
- Designed** ETL workflows orchestrated with **Apache Airflow** and stored processed data in **PostgreSQL**
- Implemented scalable **data ingestion** and **transformation pipelines** for real-time analytics

YouthLink Web Platform | Python, Flask, SQLite, HTML, CSS, Bootstrap

- Developed a **full-stack Flask web platform** enabling users to browse internships, scholarships, and opportunities
- Built **database-backed search and document** management features using **SQLite**
- Designed a **responsive user interface** with **Bootstrap** to improve accessibility across devices

EXTRACURRICULAR ACTIVITIES

Resident Assistant

February 2025 – Present

- Mentored **20+ students** and coordinated **community-building events**, improving student engagement by **60%**